

# Automated search recovers exact rank-7 matrix multiplication

Dylan Lawless<sup>1,2,3</sup>

<sup>1</sup>Quantitative Omics Epidemiology Group

<sup>2</sup>University of Zurich, Zurich, Switzerland

<sup>3</sup>Switzerland Omics, Zurich, Switzerland

7 October 2026

## Abstract

Strassen’s rank-7 algorithm for  $2 \times 2$  matrix multiplication gives recursive arithmetic exponent  $\log_2 7 \simeq 2.80735$ , while recent work proves the asymptotic bound  $\omega \leq 9/4$  over  $\mathbb{C}$ . This gap motivates the constructive search for explicit finite matrix-multiplication algorithms. We present `fast_mm_search`, a numerical-to-exact procedure that searches for bilinear decompositions from random coefficients, reconstructs rational candidates, and verifies them using exact arithmetic. With only the  $2 \times 2$ , rank-7 problem specified, and without supplying Strassen’s coefficients, the search recovered an integer-coefficient seven-product identity whose factorisation differs from the standard Strassen formula. Exact rational verification matched all 64 coefficients of the multiplication tensor. The result demonstrates that a nontrivial exact finite matrix-multiplication algorithm can be recovered automatically and establishes the procedure for searches at larger matrix orders and lower recursive exponents.

## 1 Introduction

A rank- $r$  bilinear algorithm for multiplying  $u \times u$  matrices gives recursive arithmetic exponent

$$\log_u r.$$

Strassen showed that  $2 \times 2$  matrix multiplication can be computed with seven scalar products instead of the eight used by direct multiplication (1). His  $(u, r) = (2, 7)$  construction therefore gives

$$\log_2 7 \simeq 2.80735,$$

establishing that the cubic exponent of direct matrix multiplication is not optimal.

The matrix-multiplication exponent  $\omega$  describes the asymptotic arithmetic complexity of multiplying square matrices. Recent work proves

$$\omega \leq \frac{9}{4}$$

over  $\mathbb{C}$  (2). This asymptotic result establishes that substantially lower complexity is possible, while leaving a constructive question: which explicit finite bilinear decompositions can realise improved recursive exponents?

We developed `fast_mm_search` to search for finite decompositions numerically, reconstruct candidate coefficients as exact rationals, and verify the resulting identities independently. The classical  $(2, 7)$  problem provides a controlled validation because an exact solution is known while its coefficients can be withheld from the search.

## 2 Methods

For square matrices of order  $u$ , let

$$T_u \in \mathbb{Q}^{u^2 \times u^2 \times u^2}$$

denote the matrix-multiplication tensor. A rank- $r$  bilinear algorithm has the decomposition

$$\hat{T}_u = \sum_{k=1}^r x_k \otimes y_k \otimes z_k,$$

where  $x_k$  and  $y_k$  define linear forms in the entries of the two input matrices and  $z_k$  distributes each bilinear product among the output entries. The dense parameterisation contains  $3ru^2$  scalar coefficients before accounting for term-wise scaling symmetries.

The search separates numerical discovery from exact proof:

random initialisation  $\longrightarrow$  numerical optimisation  $\longrightarrow$  rational reconstruction  $\longrightarrow$  exact verification

The validation experiment used  $u = 2$  and  $r = 7$ , giving 84 raw search coefficients.

Independent random initial values were optimised with L-BFGS-B using the relative tensor residual

$$\rho = \frac{\|T_u - \hat{T}_u\|_F}{\|T_u\|_F}.$$

Candidates satisfying  $\rho < 10^{-6}$  were term-normalised and reconstructed as rational coefficients with denominator bounded by 16. A separate verifier then evaluated the complete multiplication tensor using exact rational arithmetic. For matrix order  $u$ , this checks all  $u^6$  tensor coefficients and accepts a candidate only when every coefficient agrees exactly with  $T_u$ .

### 3 Results

The blind  $(2, 7)$  search recovered an exact rank-7 integer-coefficient identity. Before exact reconstruction, the accepted numerical candidate had relative residual

$$\rho = 3.093299 \times 10^{-8}.$$

Another candidate reached the smaller residual  $1.596061 \times 10^{-8}$  but did not yield an exactly verified identity. Numerical proximity therefore identified candidates without determining exact recoverability.

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix},$$

the recovered algorithm computes

$$\begin{aligned} P_1 &= (b - d)h, & P_2 &= a(e + f + g + h), \\ P_3 &= (a - b)(g + h), & P_4 &= (a - c)f, \\ P_5 &= (c - d)e, & P_6 &= d(e + g), \\ P_7 &= (a - d)(e + g + h), \end{aligned}$$

and reconstructs

$$AB = \begin{pmatrix} -P_1 - P_3 + P_6 + P_7 & P_1 + P_2 - P_6 - P_7 \\ P_5 + P_6 & P_2 - P_4 - P_6 - P_7 \end{pmatrix}.$$

Direct expansion gives

$$AB = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}.$$

The independent verifier evaluated all

$$2^6 = 64$$

coefficients of the multiplication tensor using exact rational arithmetic. All coefficients matched, giving zero mismatches and zero exact residual.

## 4 Discussion

The recovered identity attains the classical rank-7 bound for  $2 \times 2$  matrix multiplication. It was obtained from random initial coefficients without access to Strassen’s construction, and its displayed bilinear products differ from the standard presentation of Strassen’s formula. The result therefore establishes automated recovery and exact verification of a nontrivial finite matrix-multiplication algorithm; it does not claim a new rank bound.

Numerical optimisation and exact verification answer different questions. The smallest floating-point residual observed in the campaign did not produce the exactly verified identity, showing that residual magnitude alone cannot establish a bilinear algorithm. Rational reconstruction followed by coefficient-wise verification converts a numerical candidate into an exact tensor equality.

For a finite square decomposition, recursive complexity improves on Strassen whenever

$$\log_u r < \log_2 7,$$

equivalently

$$r < u^{\log_2 7}.$$

At  $u = 3$ ,

$$3^{\log_2 7} \simeq 21.85,$$

so  $r = 21$  is the largest integer rank at this order that would improve on recursive Strassen:

$$\log_3 21 \simeq 2.77124 < 2.80735.$$

The pair  $(3, 21)$  therefore defines a natural finite search boundary beyond the  $(2, 7)$

validation problem.

The corresponding dense parameterisation grows from

$$3 \times 7 \times 2^2 = 84$$

raw coefficients for  $(2, 7)$  to

$$3 \times 21 \times 3^2 = 567$$

for  $(3, 21)$ , while the multiplication tensor grows from

$$2^6 = 64 \quad \text{to} \quad 3^6 = 729$$

coefficients. The increase quantifies the computational step from the validation problem to a finite target whose exact solution would already cross the Strassen exponent.

A rank-21  $3 \times 3$  algorithm would give exponent approximately 2.77124, still well above the asymptotic bound  $9/4 = 2.25$ . The  $9/4$  theorem therefore supplies an asymptotic benchmark rather than a claim that a small  $(3, 21)$  decomposition must exist. The present result establishes a numerical-to-exact mechanism for searching finite rules and certifying any recovered identity independently.

## Code availability

Source code, configurations, and complete run artefacts are available at [github.com/switzerlandomics/ai-fast\\_mm\\_search](https://github.com/switzerlandomics/ai-fast_mm_search).

## References

- [1] V. Strassen, “Gaussian elimination is not optimal,” *Numerische Mathematik*, vol. 13, pp. 354–356, 1969. [doi:10.1007/BF02165411](https://doi.org/10.1007/BF02165411).
- [2] OpenAI, “An Upper Bound of  $9/4$  for the Matrix Multiplication Exponent,” October 2, 2026. <https://github.com/openai/math/tree/main/preprints/Matrix-Multiplication-Nine-Fourths-October-2-2026>.